Pregnancy Outcomes in US Prisons, 2016–2017

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Objectives. To collect national data on pregnancy frequencies and outcomes among women in US state and federal prisons.

Methods. From 2016 to 2017, we prospectively collected 12 months of pregnancy statistics from a geographically diverse sample of 22 state prison systems and the Federal Bureau of Prisons. Prisons reported numbers of pregnant women, births, miscarriages, abortions, and other outcomes.

Results. Overall, 1396 pregnant women were admitted to prisons; 3.8% of newly admitted women and 0.6% of all women were pregnant in December 2016. There were 753 live births (92% of outcomes), 46 miscarriages (6%), 11 abortions (1%), 4 stillbirths (0.5%), 3 newborn deaths, and no maternal deaths. Six percent of live births were preterm and 30% were cesarean deliveries. Distributions of outcomes varied by state.

Conclusions. Our study showed that the majority of prison pregnancies ended in live births or miscarriages. Our findings can enable policymakers, researchers, and public health practitioners to optimize health outcomes for incarcerated pregnant women and their newborns, whose health has broad sociopolitical implications. (*Am J Public Health.* 2019;109:799–805. doi:10.2105/AJPH.2019.305006)

See also Goshin and Colbert, p. 657.

At the end of 2016, there were 111 616 women in prisons across the United States, a 742% increase from the 13 258 women in prisons in 1980.[1,2] The United States has 4% of the world's female population but 30% of its female incarcerated population.[3] Although there has also been an exponential rise among men—as part of the complex political, social, racial, and public health phenomenon known as mass incarceration—the rate of increase of women in custody has outpaced that of men.[1,2,4] Nonetheless, there is a dearth of research about gender-specific health conditions among incarcerated women, especially pregnancy.

Three quarters of incarcerated women are of childbearing age (between 18 and 44 years).[2] Two thirds are mothers and the primary caregivers to young children, and up to 84% have been pregnant in the past.[5,6] In addition, up to 80% of incarcerated women report that they had been sexually active with men in the 3 months before their incarceration, and only 21% to 28% were using a reliable method of contraception.[5,7] Thus, some women will enter prison pregnant. Yet, to our knowledge, there are no systematic reports of pregnancy outcomes in US prisons.

Prison pregnancy data are critical in ensuring that incarcerated women's pregnancy-related health care needs are addressed and in helping optimize outcomes for them and their newborns. The far-reaching consequences of the health of incarcerated people for the public's health and that of broader society are well documented; these consequences are compounded for incarcerated pregnant women given that incarceration affects not only their health but also that of subsequent generations.[8–12]

Documenting pregnancy outcomes in prisons is a matter of health equity and reducing maternal health disparities. Black women are imprisoned at twice the rate of White women, a manifestation of the racism embedded in the US criminal legal system[2,13]; in addition, the preincarceration lives of a significant proportion of women in prison are characterized by poverty, substance use disorders, histories of trauma and abuse, and limited access to health care.[11,12] Incarcerated pregnant women are more likely to have these and other risk factors for poor perinatal outcomes than are nonincarcerated pregnant women.[14,15] Prisons are constitutionally required to provide health care[16]; however, no mandatory standards, oversight, or requirements for data reporting are in place. Although voluntary accreditation programs exist (e.g., the National Commission on Correctional Health Care and the American Correctional Association), this lack of standardized health services results in tremendous variability in pregnancy care in prisons.[11]

Existing prison pregnancy data are scant and outdated. According to a 2004 Bureau of Justice Statistics (BJS) survey, 3% of women in federal prisons and 4% in state prisons reported that they were pregnant at intake.[17] Much has changed in the criminal legal system since these data were collected, including 15 618 more women in prison in 2016 than in 2004[2,18]; moreover, some state and local policies have disproportionately targeted pregnant women who use drugs for incarceration.[19] Thus, one cannot assume that the pregnancy situation in prison has remained constant. Likewise, the only national estimate of births in prisons comes from a survey done in 1998, reporting 1400 births at 43 state prisons[20]; to our knowledge, there has never been a systematic assessment of abortions, stillbirths, miscarriages, ectopic pregnancies, or neonatal and maternal deaths in prisons.

Although several small studies at individual prisons and jails and a pair of systematic

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This article was accepted January 15, 2019.
doi: 10.2105/AJPH.2019.305006

reviews have reported outcomes such as preterm births, stillbirths, and low birth weight, these studies are based on retrospectively collected data, limited in terms of outcomes, outdated, and not systematic or representative in their sampling.[21–25] Further contributing to the lack of maternal health and birth data from prisons is the exclusion of incarceration from national health statistics databases. Specifically, information on mothers' incarceration status is not collected on birth certificates; no federal agency that collects birth data tracks pregnancy in prison. Moreover, no federal agency that collects incarceration statistics (namely, the BJS) records pregnancy data. To address these data gaps in pregnancy outcomes among incarcerated women, the Pregnancy in Prison Statistics study prospectively collected 1 year of pregnancy outcome data from state and federal prisons across the United States.

METHODS

From May 2016 to December 2016, state prison systems and Federal Bureau of Prisons (BOP) institutions were recruited through purposive and snowball sampling. Using demographic data published by the BJS, we targeted initial recruitment to the 18 state prison systems housing at least 2000 women. Custody and health care administrators at these state prison systems were contacted via e-mail or telephone and invited to participate. As a means of broadening the sample, networks established through the National Institute of Corrections and the National Resource Center on Justice Involved Women were used to notify prison systems housing fewer than 2000 women. BOP administrators were contacted, and the acting director of the Department of Justice at the time approved BOP institutions' participation in reporting pregnancy data.

Both state and federal prisons house people sentenced to serve terms typically longer than 1 year, so most imprisoned women will be in custody for the duration of their pregnancies. Otherwise, state and federal prisons differ in their administrative structure and oversight, geographic locations, and types of convictions. Pregnancy care at federal prisons must follow BOP policies, whereas each state prison falls under the jurisdiction of the state's department of corrections policies.

Twenty-two state prison systems and all 26 BOP prisons housing women enrolled in the study (see the box on this page; Figure A, available as a supplement to the online version of this article at http://www.ajph.org). Study reporting occurred between May 2016 and January 2018, with the exact 1-year time frame for each prison based on when it was enrolled. According to 2016 BJS prison census data, the study prisons included 57% of all women in US prisons (53% of women in state prisons and 86% of women in federal prisons).[2] In 17 of the 22 state systems, pregnant people were housed at 1 facility. In 4 states more than 1 prison housed pregnant women, and their study reporting reflected all pregnant women in the state. Wisconsin housed pregnant women at 4 locations, but data were reported for only a single prison. BOP reporting excluded outcomes among 25 women who were transferred to privately run federal prisons. Rhode Island and Vermont combine jail and prison populations (with jail populations being predominantly of pretrial status and serving shorter sentences), and midway through the study Maryland began housing pregnant pretrial women in prisons; thus, these states' data reflect both convicted and pretrial populations.

STATE PRISON RECRUITMENT AND PARTICIPATION: UNITED STATES, 2016–2017

Participating prison systems	Alabama, Arizona, Colorado, Georgia, Illinois, Iowa, Kansas, Louisiana, Maine, Maryland, Massachusetts, Minnesota, Mississippi, Ohio, Oklahoma, Pennsylvania, Rhode Island, Tennessee, Texas, Vermont, Washington, Wisconsin
Prison systems that declined	California, Connecticut, Florida, Indiana, New York, North Carolina, Oregon, Virginia
Prison systems that did not respond	Kentucky, Michigan, Missouri
Prison system that were not contacted	Alaska, Arkansas, Delaware, Hawaii, Idaho, Montana, Nebraska, Nevada, New Hampshire, New Jersey, New Mexico, North Dakota, South Carolina, South Dakota, Utah, West Virginia, Wyoming

Designated reporters at the different sites (medical directors, other health care personnel, prison research coordinators, women's programming coordinators, and, in one instance, a warden) reported data at the end of each month. We provided sites with examples of tracking systems, as the majority of the participating prisons had no existing process for tracking pregnancy data. Data were reviewed monthly by study staff to check for discrepancies; site reporters were contacted directly to resolve inconsistencies. Midway through the study, we approached all of the site reporters for an interview to ensure that we could understand their tracking system and to ensure accuracy of reporting; 43% of site reporters responded and were interviewed, and all were reporting accurately.

The secure, Web-based REDCap (research electronic data capture) application was used to collect and manage study data.[26] Site reporters could also complete electronic documents and return them via e-mail, fax, or mail. Data were collected for 13 consecutive months with the exception of Louisiana, which temporarily suspended reporting from August 2016 to February 2017 as a result of flooding and then resumed. Thirteen months of data were collected because we assumed that the first month would be a trial period for pregnancy tracking systems. Analyses were performed on data from months 2 to 13 (a total of 12 months).

Prisons reported the following aggregate, deidentified numbers each month: pregnant women admitted and in custody on the last day of the month; live births and stillbirths, stratified by term (more than 37 weeks), preterm (24 weeks to 36 weeks, 6 days), and very early preterm (20 weeks to 23 weeks, 6 days); preterm and cesarean deliveries; miscarriages; abortions; ectopic pregnancies; maternal deaths during custody (deaths during pregnancy or within 6 weeks of the pregnancy ending); and newborn deaths within the first 3 days of life. This differs from the standard definition of neonatal death within the first 30 days because prison systems could consistently know what happened to the infant only during the time the mother was in the hospital for her postpartum recovery. Information was also collected on pregnancy testing policies and on whether the prison's health care system was privatized or accredited.

As a result of the nature of aggregate data collection from each prison system, we were not able to gather information on individual women's specific demographic characteristics (e.g., race and age). We also could not assess whether any of these pregnancies occurred in transgender men or non-binary individuals. Prisons also reported their overall female census and total number of admitted women on December 31, 2016 (consistent with BJS reporting), and on July 31, 2016, so that we could calculate the proportion of imprisoned women who were pregnant and the proportion of admitted women who were pregnant at 2 points in time; July 31 census data were available only for 16 states and admission data for 13.

Data were analyzed for frequencies and other descriptive statistics. Proportions of each pregnancy outcome were calculated for pregnancies in which the outcomes were known; that is, outcomes did not account for women who were released from prison while they were still pregnant or those who were in custody but still pregnant at the end of the study reporting period. We performed analyses for all prisons combined and also broken down by state and federal jurisdiction. Because the data represent complete counts from all state prison systems (with the exception of Wisconsin), we were not able to calculate confidence intervals.

RESULTS

The participating state prisons represented a range of small, medium, and large prisons holding between 132 and 12 711 women, with most sites (54%) housing 500 to 2000 women (Table 1). Sixty-four percent of prisons had some form of voluntary health care accreditation, and 50% contracted health care delivery to a private corporation. In total, 1396 pregnant women were admitted to these state and federal prisons over the 12-month study period (with a range of 0–26 women admitted each month at individual prison systems; Table 2).

At the 13 state prisons for which data on total female admissions were available, 65 of 1646 women (3.9%) in July 2016 and 63 of 1654 (3.8%) in December 2016 were pregnant at the time of their admission (with a range of 0% to nearly 17% of admissions in individual states; Table 2). The overall pregnancy prevalence at these sites on July 31, 2016, was 0.9%. The 2016 year-end pregnancy prevalence at all state and federal prisons was 0.6% (Table 2). Among women who were already in state prisons, 5 new pregnancies were diagnosed during a 6-month reporting period. Three of these women became pregnant during work release; the timing of the other 2 pregnancies was not reported.

Overall, there were 753 live births, representing 92% of all known pregnancy outcomes (Table 3); 685 births occurred among women incarcerated in state prisons and 68 among women incarcerated in BOP sites. Among live births in state prisons, 6% were preterm and 0.3% were very early preterm. The majority of live births were vaginal, with 32% cesarean deliveries. All but 6 births occurred in a hospital; 3 were attributable to precipitous labor with prison nurses or paramedics in attendance, and details were not available for the others. Of the remaining 8% of non-live-birth pregnancy outcomes, 6% were miscarriages (n = 46), 1% were abortions (n = 11), 0.5% were stillbirths (n = 4), and 0.25% were ectopic pregnancies (n = 2). The 4 stillbirths occurred among women who delivered preterm. There were 3 newborn deaths and no maternal deaths. Table 3 presents data on selected outcomes according to state and federal jurisdiction.

More miscarriages were reported at state prisons that administered pregnancy tests within 48 hours of admission (n = 33) than at sites that conducted pregnancy tests only at the woman's request or at the clinician's discretion (n = 9 miscarriages), suggesting higher detection of early pregnancy outcomes.

TABLE 1—Characteristics of Participating US State Prisons: 2016–2017 (n = 22)

Characteristic	No. (%)
Region	
Northeast	5 (23)
Midwest	6 (27)
South	8 (36)
West	3 (14)
Overall female census, no.[a]	
< 500	3 (14)
501–1000	8 (36)
1001–2000	4 (18)
2001–5000	6 (27)
> 5000	1 (5)
Health care accreditation	
None	8 (36)
National Commission on Correctional Health Care	5 (23)
American Correctional Association	11 (50)
Privately contracted health care	11 (50)
Privately contracted prison administration	1 (5)
Pregnancy test performed on routine medical intake	14 (64)
Conjugal visits permitted	1 (5)
Furlough prior to due date	1 (5)

[a]Based on reported census on December 31, 2016.

DISCUSSION

To our knowledge, this study is the first systematic investigation to collect prospective data on pregnancy frequencies and outcomes among women in US prisons. Our study demonstrates that it is feasible to prospectively track pregnancy outcome data among incarcerated women. The vast majority of pregnancies (92%) ended in live births. Approximately 4% of women admitted to state prisons were pregnant, a percentage similar to the figure reported in the 2004 BJS survey that included self-reported pregnancy status at intake.[17] This represents a notable proportion of women who will arrive pregnant and will have pregnancy-specific health care needs that must be addressed in a timely fashion. Collection of data on pregnancy frequencies and outcomes among women in prison, who are not accounted for in existing databases, is part of a broader public health strategy of tracking maternal health data to reduce adverse outcomes and promote equity.[27]

The overall prevalence of pregnancy on December 31, 2016, was 0.7% at all state sites and 0.3% at all federal prisons. This overall pregnancy prevalence differs from the proportion of admitted women who

TABLE 2—Pregnancy Prevalence and Admissions of Pregnant Women in US Prisons, by State and Federal Jurisdiction, 2016–2017

Prison System	Total Female Census, December 31, 2016	Pregnancy Prevalence, December 31, 2016 (All Women, %)	Total Female Census, July 31, 2016	Pregnancy Prevalence, July 31, 2016 (All Women, %)	Total No. of Admissions of Pregnant Women (Over 12 Months)	Admitted Women Who Were Pregnant, %, December 2016	Admitted Women Who Were Pregnant, %, July 2016	Admissions of Pregnant Women per Month, Median (Range)	Pregnant Women in Prison on Last Day of Month, Median (Range)
Alabama[a]	885	1.5	1 170	1.0	36	2.3	5.3	3 (0–)	9 (5–13)
Arizona	3 978	0.5	4 035	0.8	92	3.8	2.1	8.5 (3–12)	25 (18–31)
Colorado	1 564	0.8	1 858	0.9	39	3.7	7.4	3.5 (0–7)	14 (9–17)
Georgia[b]	3 788	0.6	3 739	0.6	85	. . .	. . .	7 (4–10)	21 (17–27)
Illinois	2 580	0.3	. . .	. . .	35	. . .	. . .	3 (0–9)	10 (6–17)
Iowa	689	1.2	693	1.9	22	3.1	0.0	2 (0–5)	8.5 (7–15)
Kansas	843	0.8	. . .	. . .	33	. . .	. . .	3 (0–6)	7.5 (4–11)
Louisiana	641	0.8	915	0.9	24	5.4	7.6	1.5 (0–6)	4.5 (2–8)
Maine	235	0.4	200	0.0	5	0.0	0.0	0 (0–2)	0.5 (0–3)
Maryland	750	1.1	779	1.2	36	16.7	6.3	2.5 (0–6)	9 (6–10)
Massachusetts	571	2.5	512	2.0	95	4.7	6.1	8.5 (3–14)	10.5 (6–14)
Minnesota	662	2.1	. . .	. . .	49	. . .	. . .	4 (2–7)	14 (10–15)
Mississippi	1 369	0.4	1 348	0.6	33	. . .	. . .	2 (0–8)	7 (3–11)
Ohio	4 594	1.0	2 618	2.0	138	. . .	. . .	12.5 (4–24)	42.5 (27–55)
Oklahoma	3 006	0.5	. . .	0.6	44	2.0	4.0	3.5 (0–6)	16 (7–20)
Pennsylvania	2 599	0.5	. . .	. . .	43	. . .	. . .	3 (1–8)	16.5 (9–26)
Rhode Island	132	3.8	133	0.8	47	7.5	5.7	4 (1–8)	1 (0–5)
Tennessee	1 785	0.2	2 915	0.3	29	1.5	1.0	2 (0–5)	7 (2–11)
Texas	12 434	0.5	. . .	. . .	241	. . .	. . .	19.5 (13–26)	61 (54–78)
Vermont	135	4.4	. . .	. . .	25	. . .	. . .	1 (0–5)	2 (0–6)
Washington	1 275	1.2	1 367	1.1	40	4.5	5.1	4 (0–5)	14.5 (5–17)
Wisconsin (Taycheedah[c])	851	0.8	878	0.9	33	4.6	2.6	2 (0–6)	7.5 (4–10)
Total, state prisons	45 366	0.7	26 203	0.9	1 224	3.8	3.9	3 (0–26)	9.5 (0–78)
Total, federal prisons	10 896	0.3	. . .	. . .	172	. . .	. . .	15.5 (7–21)	37.5 (33–47)
Total, all prisons	56 262	0.6	. . .	. . .	1 396	. . .	. . .	3.5 (0–26)	10 (0–78)

[a]Female admission and census data were obtained from publicly available reports (http://www.doc.state.al.us/StatReports).
[b]Female census data for July 2016 were obtained from publicly available reports (http://www.dcor.state.ga.us/Research/Monthly_Profile_all_inmates).
[c]Wisconsin housed pregnant women at 4 locations, but data were reported for only 1 prison.

were pregnant because the former encompasses in the denominator all women in prison, including those who have been there for many years and are less likely to become pregnant because of their confinement and age; women newly admitted to prison, in contrast, were more recently in the community, where they may have become pregnant.

It is notable that there was significant variation by state. For instance, Texas and Ohio had some months when there were more than 50 pregnant women in their prisons on the last day of the month, whereas other states had no incarcerated pregnant women; this is likely related to Texas having the largest and Ohio the fourth largest populations of incarcerated women in the United States.[2] Likewise, the percentages of newly admitted women who were pregnant in July and December varied widely by state, with percentages as high as 17% in Maryland and as low as 0% in Maine. Of note, Maryland's July to December 2016 increase in the proportion of admitted pregnant women was probably attributable to the previously mentioned policy change according to which some pretrial pregnant women were housed at the state prison. Overall, miscarriages represented 6% of known pregnancy outcomes; these percentages also varied by state, however,

TABLE 3—Selected Pregnancy Outcomes in US Prisons, by State and Federal Jurisdiction, 2016–2017

Prison System	Total Female Census, December 31, 2016	Total Live Births, No. (% of Known Outcomes)	Total Miscarriages, No. (% of Known Outcomes)	Preterm Births, No. (% of Live Births)	Cesarean Deliveries, No. (% of Live Births)
Alabama	885	21 (95)	1 (5)	0 (0)	11 (52)
Arizona	3 978	28 (80)	7 (20)	0 (0)	8 (29)
Colorado	1 564	25 (89)	2 (7)	2 (8)	6 (24)
Georgia	3 788	50 (91)	4 (7)	0 (0)	18 (36)
Illinois	2 580	30 (88)	4 (12)	1 (3)	13 (43)
Iowa	689	16 (94)	0 (0)	1 (6)	3 (19)
Kansas	843	11 (79)	3 (22)	0 (0)	0 (0)
Louisiana	641	13 (93)	0 (0)	1 (8)	5 (38)
Maine	235	2 (100)	0 (0)	0 (0)	0 (0)
Maryland	750	18 (95)	0 (0)	0 (0)	4 (22)
Massachusetts	571	8 (67)	2 (17)	2 (25)	5 (63)
Minnesota	662	21 (81)	5 (19)	0 (0)	10 (48)
Mississippi	1 369	15 (100)	0 (0)	0 (0)	4 (27)
Ohio	4 594	101 (92)	7 (6)	16 (16)	22 (22)
Oklahoma	3 006	36 (100)	0 (0)	1 (3)	14 (39)
Pennsylvania	2 599	32 (97)	1 (3)	1 (3)	6 (19)
Rhode Island	132	0 (0)	0 (0)	0 (0)	0 (0)
Tennessee	1 785	26 (100)	0 (0)	1 (4)	15 (58)
Texas	12 434	171 (97)	5 (3)	10 (6)	61 (36)
Vermont	135	1 (25)	1 (25)	0 (0)	1 (100)
Washington	1 275	36 (95)	0 (0)	2 (6)	7 (19)
Wisconsin (Taycheedah[a])	851	24 (100)	0 (0)	1 (4)	8 (33)
Total, state prisons	45 366	685 (92)	42 (6)	39 (6)	221 (32)
Total, federal prisons	10 896	68 (91)	4 (5)	. . .	. . .
Total, all prisons	56 262	753 (92)	46 (6)	. . .	. . .

[a]Wisconsin housed pregnant women at 4 locations, but data were reported for only 1 prison.

with miscarriages representing 20% or more of pregnancy outcomes in some states.

Numerous factors likely contribute to these state-by-state variations in pregnancy frequency and miscarriage, including state sentencing laws, prison health care policies, community reproductive health care access, individual patient attributes, and other variables that our study did not measure.

It is useful to compare our study results with national pregnancy statistics for the general population; however, such a comparison is limited by differences in the ways in which national statistics are collected and our study statistics were gathered. The National Survey of Family Growth (2011–2013) reported that 5% of women 15 to 44 years old were pregnant or postpartum at the time of their interview.[28] To estimate a proportion of imprisoned pregnant women from our study with a similar age denominator, we can assume, on the basis of BJS data, that 75% of all women at Pregnancy in Prison Statistics study sites were 18 to 44 years old in 2016[2]; this yields a pregnancy prevalence of 0.8%. The national percentage was higher than our study's prison pregnancy preva-lence; however, our study did not include postpartum women in the denominator.

The general fertility rate, a measure of the number of live births among women 15 to 44 years of age, was 62 per 1000 women in the United States in 2016.[29] For our study data, we can roughly estimate a general fertility rate of 18 per 1000 imprisoned women 18 to 44 years old. Although this estimate is understandably lower than national rates, it is still notable, and it speaks to the need to address the numerous complexities of birth in custody, such as the medically unsafe practices of placing pregnant women in solitary confinement and shackling women in labor, ensuring proper pregnancy and postpartum care, and determining who will care for the infants born to mothers in custody.[11,30]

Nationally, nearly 10% of live births in 2016 were preterm,[29] as compared with 6% of live births in prisons in our study. This lower preterm birth proportion in these prisons may be partially related to the relative presence of prenatal care, food, and shelter and the limited access to illicit substances, conditions that may be different for some pregnant women not in custody; however, this explanation should be considered cautiously given the variability in access to and quality of prenatal care from prison to prison. In addition, some state prison systems had preterm birth rates that exceeded the national rate, suggesting that the context of the individual prison system and pre-incarceration conditions may play a role.

Our relative preterm birth rates should further be considered with caution given the problematic implication that prisons—designed for punishment, not health care—may somehow exert a "protective" effect on pregnancy. Such an interpretation fails to consider the broader social and structural determinants of health that differentially affect people before, during, and after incarceration.[12]

Limitations

Our study has several limitations. We could not assess gestational age at prison entry, and variations in gestational age may have been correlated with the incidence of various outcomes. It is also possible that selection bias influenced pregnancy outcomes, as prisons that chose to participate in this study may already be more attuned to addressing the needs of pregnant incarcerated women. We could not collect data on individual-level details such as women's race, socioeconomic status, preincarceration health, or prior pregnancy history, all factors that likely influenced these women's pregnancy outcomes.

Although there may be trends in certain outcomes by state, prison size, and pregnancy testing policies, we cannot make assessments of associations because of the high degree of variability of conditions at different state prisons (e.g., the type of hospital where imprisoned women delivered), which we could not take into account. For instance, small prisons such as those in Vermont and Rhode Island had the highest pregnancy prevalence, but both of these states have unified jail and prison systems that house pretrial individuals. The experiences of pregnant women in prisons may be different in states with small numbers of these women than in states that are more accustomed to having larger numbers of pregnant women. Thus, we were limited in our ability to make state-by-state comparisons.

We captured data from prisons housing 57% of imprisoned women in the United States. Because of resource limitations, we could not collect pregnancy statistics from 28 state prison systems, including 3 large systems that declined to participate (California, Florida, and New York). It is difficult to predict what the trends would be if all 50 states were included. If patterns in the entire population of women in prison were comparable to those we analyzed, we could proportionally extrapolate that, in 2016, nearly 2500 pregnant women were admitted to state and federal prisons, resulting in 1300 births. However, because our data show tremendous variability from state to state, it is unlikely that patterns in nonincluded prisons were similar to those in our study prisons, and this estimate is extremely tentative. This is precisely why a more comprehensive set of prisons should be studied.

It is also important to consider the implications of our data being deidentified and aggregated. Each statistic represents a range of experiences of individual pregnant incarcerated women. Being in prison or jail during pregnancy can be a difficult time for many women, fraught with uncertainty about the kind of health care they might receive, about whether they will be shackled in labor, and about what will happen to their infants when they are born. Some pregnant women in custody may experience isolation and degradation from staff and insufficient prenatal care.[10,11,20] Further research is warranted to identify and disaggregate the individualized experiences of women who are pregnant and behind bars to better understand their needs and the contexts in which their pregnancies unfold.

Public Health and Social Justice Implications

Our data make important contributions to an area in which there is substantial neglect. That prison pregnancy data have previously not been systematically collected or reported signals a glaring disregard for the health and well-being of incarcerated pregnant women. The BJS collects data on deaths during custody[31] but not births during custody. Despite this marginalization, it is important to recognize that incarcerated women are still members of broader society, that most of them will be released, and that some will give birth while in custody; therefore, their pregnancies must be counted.

Given mass incarceration's racialized dimensions and the fact that imprisoned women are disproportionately women of color,[2] understanding what happens to their pregnancies is a crucial part of broader public health efforts to understand systemic racism's impact on trenchant disparities in maternal health and pregnancy outcomes. Information about imprisoned women's pregnancies can also help improve outcomes for mothers and their children beyond pregnancy. The majority of women who give birth while in custody will be separated from their newborns soon after delivery, which imposes significant limitations on breastfeeding, bonding, and parental rights.[11]

Further research in this area is essential to track data from prisons, jails, juvenile and immigrant detention centers, and other institutions of incarceration in all 50 states. It is encouraging that federal legislation introduced in the US House of Representatives in September 2018 includes a provision that would require the BJS to collect pregnancy data from federal, state, and local institutions of incarceration.[32] Data from our study can be used to develop national standards of care for incarcerated pregnant women, advocate for policies and legislation that ensure adequate and safe pregnancy care and childbirth, develop alternatives to incarceration for pregnant women, promote reproductive justice, and encourage broader attention to the reproductive health needs of marginalized women and their families. AJPH

CONTRIBUTORS

C. Sufrin originated the study, contributed to data collection, led the data analysis and interpretation, and led the writing and preparation of the article. L. Beal coordinated data collection, co-led the data analysis, and assisted with the writing and preparation of the article. J. Clarke contributed to study design, data interpretation, and the writing of the article. R. Jones and W. D. Mosher contributed to study design, data analysis and interpretation, and the writing of the article.

ACKNOWLEDGMENTS

This research was supported by the Society of Family Planning Research Fund and the National Institute of Child Health and Development (grant NICHD-K12HD085845).

We are grateful to Becki Ney from the National Resource Center on Justice Involved Women and Maureen Buell from the National Institute of Corrections for their assistance in recruitment. Also, we acknowledge the site reporters at the participating prisons and the incarcerated women whose pregnancies are represented in these statistics.

Note. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

CONFLICTS OF INTEREST

The authors have no conflicts of interest to report.

HUMAN PARTICIPANT PROTECTION

This study was deemed non–human participant research by the institutional review board of the Johns Hopkins School of Medicine, and we followed each prison's system for research approval.

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